

# Data preparation I:

## Data representation and Data cleaning

Mario Martin

Universitat Politècnica de Catalunya

*[mmartin@cs.upc.edu](mailto:mmartin@cs.upc.edu)*

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# Overview

# **Data: The central element in Data mining**

## Subsection 1

# Data preparation

# Data: The central element in Data mining

- When facing a data mining problem, a great effort has to be done in the preparation of your data for the machine learning algorithms.
- Three main tasks:
  - ➊ Collection/Selection of data
  - ➋ Data preparation
  - ➌ Data understanding
- Acquisition of data usually involves database, scrapping or retrieval techniques needed to obtain the data.
- We will focus on how to pre-process the raw data obtained from these techniques.

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- Data can be...
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  - ▶ ...
- Problems have not empty intersection

## Subsection 2

### **Tables**



# Tabular form

- Table example:

| Acme Investors Incorporated |              |                |                    |               |     |       |                     |               |
|-----------------------------|--------------|----------------|--------------------|---------------|-----|-------|---------------------|---------------|
| Customer ID                 | Account Type | Margin Account | Transaction Method | Trades/ Month | Sex | Age   | Favorite Recreation | Annual Income |
| 1005                        | Joint        | No             | Online             | 12.5          | F   | 30–39 | Tennis              | 40–59K        |
| 1013                        | Custodial    | No             | Broker             | 0.5           | F   | 50–59 | Skiing              | 80–99K        |
| 1245                        | Joint        | No             | Online             | 3.6           | M   | 20–29 | Golf                | 20–39K        |
| 2110                        | Individual   | Yes            | Broker             | 22.3          | M   | 30–39 | Fishing             | 40–59K        |
| 1001                        | Individual   | Yes            | Online             | 5.0           | M   | 40–49 | Golf                | 60–79K        |

- Each row an *instance/observation/example*
- Each column a *feature/descriptor/attribute*

# Tabular form

- Table example:

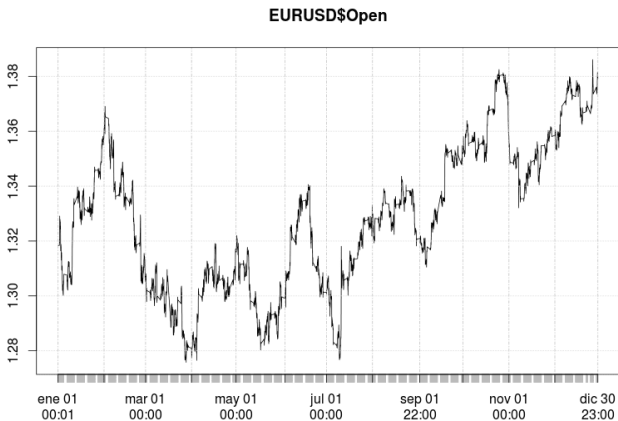
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- Each row an *instance/observation/example*
- Each column a *feature/descriptor/attribute*
- Standard form of data for Data Mining algorithms

- Usually data is extracted from relational databases. In this case data is represented in tabular form.
- But there are other sources of data like temporal series, images, transactions, textual documents that are not represented as tables.
- In this case we have to transform this information to tabular form.

# Temporal series

- Example of Temporal Data: evolution of EUR vs. USD

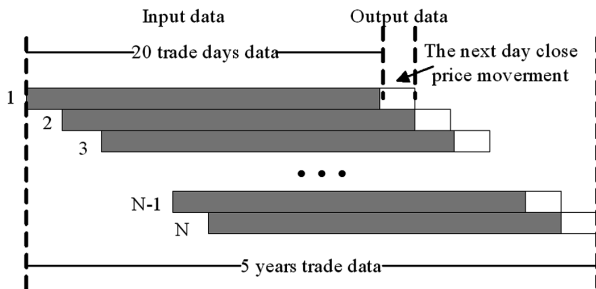


- Goal: Tabular representation to predict next day movement

# Temporal series

- Sliding-window technique:

- 1 Choose size of window  $N$  (temporal context)
- 2 Generate rows of the table by sliding windows from beginning of temporal series to the end.

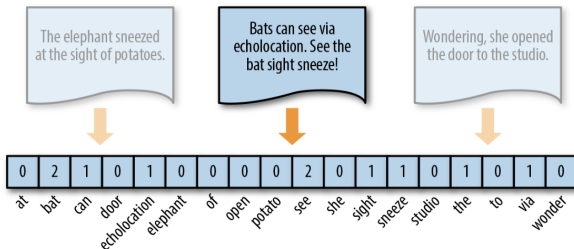


- Processing of documents is key in some data mining techniques.
  - ① Categorization of documents
  - ② Input to other DM tasks like trading
- A lot of information in textual form (webs, tweets, blogs, news,...)
- How to represent documents in tabular form?

- Processing of documents is key in some data mining techniques.
  - ① Categorization of documents
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- A lot of information in textual form (webs, tweets, blogs, news,...)
- How to represent documents in tabular form?
- Essence of document is the words that contain

# Textual Data

- Bag of word technique.
  - 1 Select a dictionary of words that will play the role of features of the table
  - 2 For each document create a row where values are the number of times each word appear in document





- In this representation we can have a large number of columns.
- Moreover, resulting tables are very sparse.

- In this representation we can have a large number of columns.
- Moreover, resulting tables are very sparse.
- In Text mining, there are ways to reduce the number of columns.
  - ▶ lemmatization
  - ▶ removal of stop-words
  - ▶ ...
- In some cases use of sparse representation of the table.

# Transactional data

- Same technique for other kind of data, for instance for transactional data:

|               |                                                                                                                                                                                                                                                                                                                                         |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Transaction 1 |     |
| Transaction 2 |                                                                                      |
| Transaction 3 |                                                                                                                                                                       |
| Transaction 4 |                                                                                                                                                                       |
| Transaction 5 |     |
| Transaction 6 |                                                                                      |
| Transaction 7 |                                                                                                                                                                       |
| Transaction 8 |                                                                                                                                                                       |

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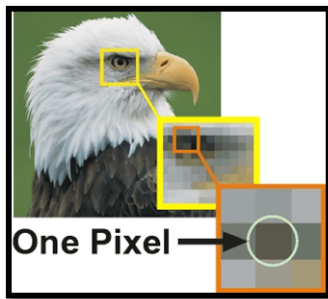
|               |                                                                                                                                                                                                                                                                                                                                         |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Transaction 1 |     |
| Transaction 2 |                                                                                      |
| Transaction 3 |                                                                                                                                                                       |
| Transaction 4 |                                                                                                                                                                       |
| Transaction 5 |     |
| Transaction 6 |                                                                                      |
| Transaction 7 |                                                                                                                                                                       |
| Transaction 8 |                                                                                                                                                                       |



|               |  |  |  |  |  |  |
|---------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Transaction 1 | 1                                                                                 | 1                                                                                  | 0                                                                                   | 1                                                                                   | 1                                                                                   | 0                                                                                   |
| Transaction 2 | 1                                                                                 | 1                                                                                  | 0                                                                                   | 1                                                                                   | 0                                                                                   | 0                                                                                   |
| Transaction 3 | 1                                                                                 | 1                                                                                  | 0                                                                                   | 0                                                                                   | 0                                                                                   | 0                                                                                   |
| Transaction 4 | 1                                                                                 | 0                                                                                  | 1                                                                                   | 0                                                                                   | 0                                                                                   | 0                                                                                   |
| Transaction 5 | 0                                                                                 | 1                                                                                  | 0                                                                                   | 1                                                                                   | 1                                                                                   | 1                                                                                   |
| Transaction 6 | 0                                                                                 | 1                                                                                  | 0                                                                                   | 1                                                                                   | 0                                                                                   | 1                                                                                   |
| Transaction 7 | 0                                                                                 | 1                                                                                  | 0                                                                                   | 0                                                                                   | 0                                                                                   | 1                                                                                   |
| Transaction 8 | 0                                                                                 | 0                                                                                  | 1                                                                                   | 0                                                                                   | 0                                                                                   | 1                                                                                   |

# Images

- Images are build from pixels



# Images

- Any image can be translated to numbers, one for each pixel, expressing the degree of intensity.



|    |    |    |    |    |   |
|----|----|----|----|----|---|
| 0  | 0  | 0  | 0  | 0  | 0 |
| 02 | 08 | 0  | 03 | 06 | 0 |
| 02 | 09 | 0  | 03 | 08 | 0 |
| 03 | 08 | 07 | 08 | 09 | 0 |
| 0  | 0  | 0  | 02 | 08 | 0 |
| 0  | 0  | 0  | 02 | 02 | 0 |

- Pixels can be used as features to describe an image in a row of a table.

|   | label | pixel0 | pixel1 | pixel2 | pixel3 | pixel4 | pixel5 | pixel6 | pixel7 | pixel8 | ... | pixel774 |
|---|-------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-----|----------|
| 0 | 1     | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | ... | 0        |
| 1 | 0     | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | ... | 0        |
| 2 | 1     | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | ... | 0        |
| 3 | 4     | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      | ... | 0        |

# Understanding your data



# Tabular form

- Table example:

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- Kind of data you can find in a table: Integers, Reals categories, booleans, ranges,...

- We categorize data into:
  - ① Numerical:
    - ① Continuous
    - ② Discrete
  - ② Categorical: Categories are "Strings"
    - ① Nominal: Not having any particular order (f.i. Color). [Include boolean features]
    - ② Ordinal: Categories have a meaningful order (rankings, grades, f.i. Age: Young,Adult,Old)

# Numerical features

- Adequate for all data mining algorithm.
- Useful information: Valid values and units.
- Common summary statistics: *Average, Std, Mean, Mode, Max, Min, boxplot...*
- Have a particular distribution that can be checked using *histograms, probability density estimation*, etc.
- When analyzing **pairs of numerical features** use *scatter plots* to visualize dependencies and *correlation* measures to summarize them.
- When comparing with **categorical feature** use summary statistics for the numerical feature for each category and compare them.

# Categorical features

- Not suited for algorithms that need to compute distances... so usually are translated to numeric values.
- Common summary statistics: *Number of different categories, Frequency of each category, Most frequent category, ...*
- Particular distribution can be checked using *histograms*.
- When analyzing **pairs of categorical features** use *heat maps* and *correlation* measures to summarize them.
- When comparing with **categorical feature** use summary statistics for the numerical feature for each category and compare them.

## Subsection 1

### **Analysis of columns**

# Analysis of columns: Uni-variate

- First step consist in *understanding* your variables, so you have to study them.
- Uni-variate analysis of a column consist in finding basic statistic description of each variable.
- For quantitative variables, max, min, median, mode, standard deviation, distribution (histograms), (boxplot), etc.
- For qualitative variables: Number of modalities, histogram.
- Useful to detect outliers, errors, etc.

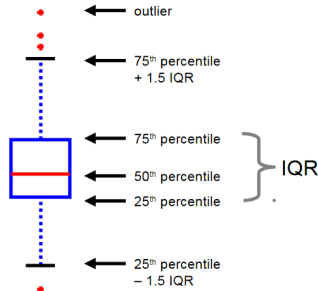
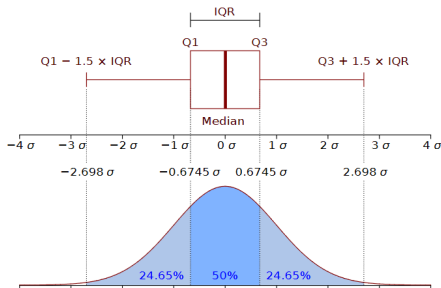
# Uni-variate analysis: statistical summary

```
>>> data = pandas.read_csv('examples/brain_size.csv', sep=';', na_values=".")
>>> data
   Unnamed: 0  Gender  FSIQ  VIQ  PIQ  Weight  Height  MRI_Count
0           1  Female  133  132  124    118    64.5    816932
1           2   Male   140  150  124    NaN    72.5    1001121
2           3   Male   139  123  150    143    73.3    1038437
3           4   Male   133  129  128    172    68.8    965353
>>> data.describe(include='all')
```

|        | Gender | FSIQ    | VIQ     | PIQ     | Weight  | Height | MRI_Count |
|--------|--------|---------|---------|---------|---------|--------|-----------|
| count  | 40     | 40.000  | 40.000  | 40.000  | 38.000  | 39.000 | 4.000e+01 |
| unique | 2      | NaN     | NaN     | NaN     | NaN     | NaN    | NaN       |
| top    | Male   | NaN     | NaN     | NaN     | NaN     | NaN    | NaN       |
| freq   | 20     | NaN     | NaN     | NaN     | NaN     | NaN    | NaN       |
| mean   | NaN    | 117.450 | 112.350 | 111.025 | 151.053 | 68.526 | 9.088e+05 |
| std    | NaN    | 31.137  | 23.616  | 22.471  | 23.479  | 3.995  | 7.228e+04 |
| min    | NaN    | 77.000  | 71.000  | 72.000  | 106.000 | 62.000 | 7.906e+05 |
| 25%    | NaN    | 90.750  | 90.000  | 88.250  | 135.250 | 66.000 | 8.559e+05 |
| 50%    | NaN    | 131.000 | 113.000 | 115.000 | 146.500 | 68.000 | 9.054e+05 |
| 75%    | NaN    | 138.250 | 129.750 | 128.000 | 172.000 | 70.500 | 9.501e+05 |
| max    | NaN    | 240.000 | 150.000 | 150.000 | 192.000 | 77.000 | 1.080e+06 |

# Uni-variate analysis: Boxplot

- Way to display information about distribution of data:

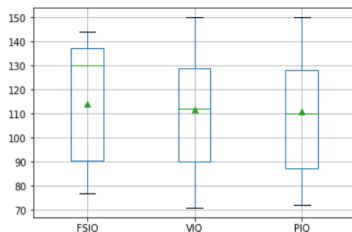




# Uni-variate: Boxplot

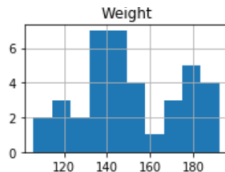
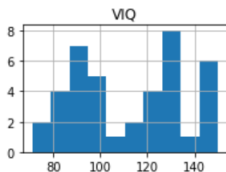
```
In [8]: new.boxplot(column=['FSIQ', 'VIQ', 'PIQ'], meanline=False, showmeans=True, return_type='dict')
print("Boxplots")
data=new
```

Boxplots



# Uni-variate: Histogram

- Another way to display information about distribution of a single variable.
- Divide the values into bins and show a bar plot of the number of objects in each bin.
- The height of each bar indicates the number of objects.
- Shape of histogram depends on the number of bins.
- Example:



# Analysis of columns: bi-variate

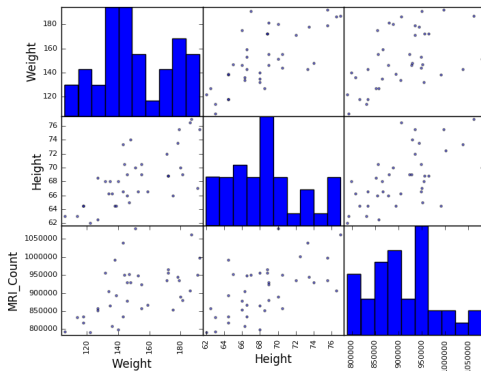
- Consist in plotting one variable against another
- Useful to detect correlations, anti-correlations, redundancies, and so understand your data

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- If both variables are quantitative, **scatter plots**, regression and correlation value

# Bi-variate analysis: Scatter plots

```
>>> from pandas.tools import plotting  
>>> plotting.scatter_matrix(data[['Weight', 'Height', 'MRI_Count']])
```



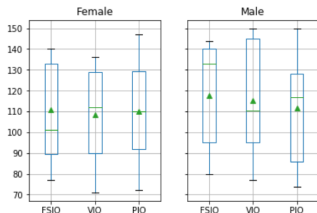
# Analysis of columns: bi-variate

- Consist in plotting one variable against another
- Useful to detect correlations, anti-correlations, redundancies, and so understand your data
- If both variables are quantitative, **scatter plots**, regression and correlation value
- If one variable is qualitative and the other quantitative, **boxplot for each modality**

# Bi-variate analysis: Boxplot for each category

```
groupby_gender.boxplot(column=['FSIQ', 'VIQ', 'PIQ'],meanline=False,showmeans=True,return_type='dict')  
print('Boxplots separated by Gender')
```

Boxplots separated by Gender



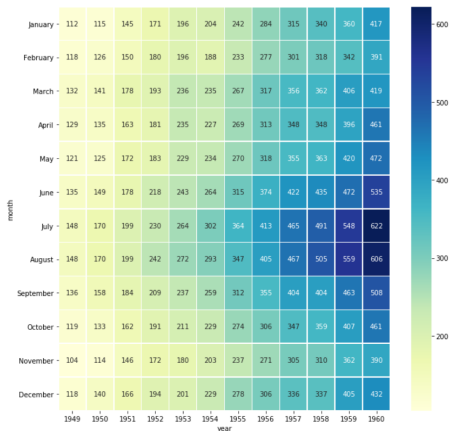
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- If both variables are quantitative, **scatter plots**, regression and correlation value
- If one variable is qualitative and the other quantitative, **boxplot for each modality**
- If both variables are qualitative, **heatmap** of each pair of modalities and  $\chi^2$  comparison



# Bi-variate analysis: heatmap

```
>>> flights = sns.load_dataset("flights")  
>>> flights = flights.pivot("month", "year", "passengers")  
>>> ax = sns.heatmap(flights)
```



# Analysis of columns: bi-variate

- Consist in plotting one variable against another
- Useful to detect correlations, anti-correlations, redundancies, and so understand your data
- If both variables are quantitative, **scatter plots**, regression and correlation value
- If one variable is qualitative and the other quantitative, **boxplot for each modality**
- If both variables are qualitative, **heatmap** of each pair of modalities and  $\chi^2$  comparison
- Theoretically this can be extended to more than bi-variate analysis, but number of comparisons grows exponentially with number of columns compared and also become harder to interpret and visualize

# Data Preprocessing

# Data Preprocessing

- Real world data are generally:
  - ▶ Incomplete: lacking attribute values, lacking certain attributes of interest.
  - ▶ Noisy: containing errors, inconsistencies or outliers.
- Tasks in data preprocessing:
  - ▶ Data cleaning: fill in missing values, remove outliers, and resolve inconsistencies.
  - ▶ Data transformation: normalization and distribution transformation of features.
  - ▶ Data reduction: reducing the features or rows but producing the same or even better results.
  - ▶ Data augmentation: augmenting the number of columns using multiple databases or building new features

## Subsection 1

# Data Cleaning I: Missing values

# Missing data

- Missing data is when for one column and row we don't have any value.
- Several possible causes:
  - ▶ Lost value in reading of data
  - ▶ Lost value in transmission of data
  - ▶ Feature not applicable to individual
- Most Data Mining algorithms need full tables to work with.
- Several solutions
  - ▶ Removal: Remove row/column with missing data
  - ▶ Imputation: Fill the missing value with something that makes sense

# Missing data: Removal

- Removal: Remove row/column with missing data
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# Missing data: Removal

- Removal: Remove row/column with missing data
- Removal of missing rows or columns can reduce a lot the amount of data available for DM algorithms.
- Even worse: Removal of rows with missing data can *bias* the dataset.
- *Only applicable when removal does not decrease the quality of the data set.*
- Applicable when missing data is concentrated in few columns or rows.
- Not recommended. In case you do that, try also applying other techniques.

# Missing data: Imputation

- In general imputation is recommended
- But how to decide the value used to fill the missing data position in table?

# Missing data: Imputation

- In general imputation is recommended
- But how to decide the value used to fill the missing data position in table?
- Use common sense. Any imputation we make should not bias the dataset.
- Several approaches.

# Missing data: Imputation

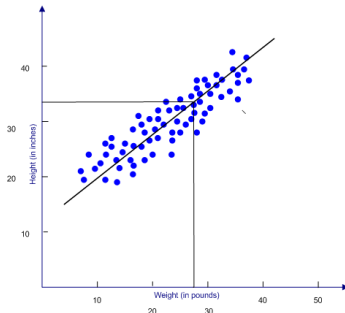
- ❶ Replace with constant global value:
  - ▶ Extreme value: For instance -1 when values are positive
  - ▶ Add constant value that makes sense
  - ▶ Add special column marking not value in other column
- ❷ Replace with feature mean/median/mode: compute the mean (numerical) or most common category (categorical) of the feature and replace the missing values of that feature with that value.
- ❸ Replace with feature mean/median/mode of the category: when DM goal is classification, substitute with mean value of the feature for the class the row belongs to<sup>1</sup>.

---

<sup>1</sup>caveat: what to do with test data with missing values?

# Missing data: Imputation

- ④ Find correlations with other features and replace with values according to this correlation



- ⑤ Substitute missing value according to the  $k$  most similar cases: (1) Apply distance function to find  $k$  closest instances. (2) Compute mean/mode/median of the  $k$  cases for missing variable and (3) replace missing value with it.

## Subsection 2

### **Finding errors**

# Finding errors

- Errors consist in data that does not corresponds to reality. Different possible causes (in measure, transcription, etc.).

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- Errors consist in data that does not corresponds to reality. Different possible causes (in measure, transcription, etc.).
- Easy to detect wrong values when they are impossible (f.i. negative price, too high age, etc.)
- This kind of errors can be detected because they are out of range, so they can be detected using uni-variate analysis of columns



# Finding errors

- Other erroneous values can be detected because they are unlikely considering the information of another correlated variable (f.i. age considering also number of descendants)
- This kind of errors can be detected because they are out of the regression line for both variables, so they can be detected using bi-variate analysis of correlated columns

# Finding errors

- Other errors are hard to find because the wrong value is plausible (so don't expect to obtain a completely clean dataset: your algorithms will have to deal with noisy data.)
- When an error is found we can do the following:
  - ❶ Nothing
  - ❷ Remove the row or the column with the error (specially when there are a lot of errors in row or column)
  - ❸ Replace the value using missing data techniques

## Subsection 3

### **Finding outliers**

# Finding outliers

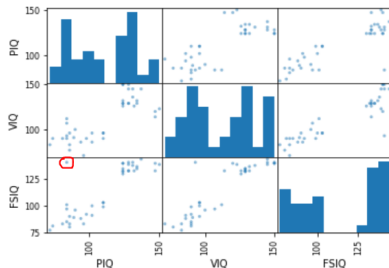
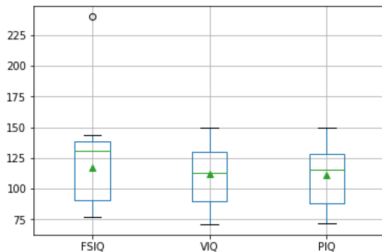
- Outliers consist in right data so different from normality that seems an error
- Detection can be done using same tools like error detection.  
But, what to do with them? You should keep them, because they are actual data, but...
- ... some data mining algorithms can be fooled because of this outliers (this also happens with basics statistic measures, for instance the mean / mode measures)
- So, in most cases is better to remove the row (be cautious)

# Finding outliers

- Working assumption: There are considerably more “normal” observations than “abnormal” observations (outliers/anomalies) in the data
- General Steps:
  - ▶ Build a profile of the “normal” behavior: Profile can be patterns or summary statistics for the overall population
  - ▶ Use the “normal” profile to detect outliers: Outliers are observations whose characteristics differ significantly from the normal profile
- Types of outlier’s detection schemes:
  - 1 Graphical
  - 2 Statistical-based
  - 3 Distance-based
  - 4 Density-based

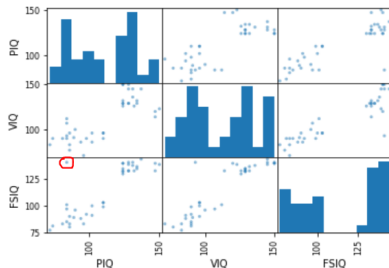
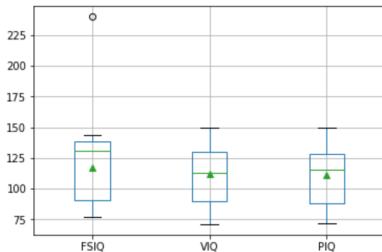
# Finding outliers: Graphical

- Generate and visualize boxplot (1-D) and Scatter plots (2-D).



# Finding outliers: Graphical

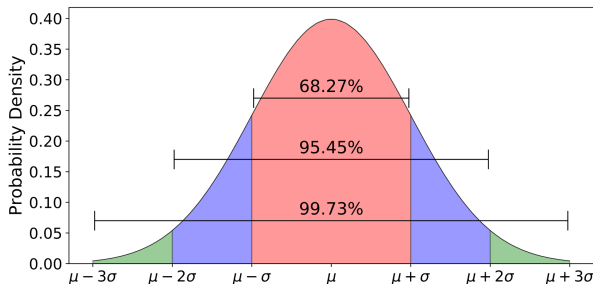
- Generate and visualize boxplot (1-D) and Scatter plots (2-D).



- Problems: Time consuming and subjective

# Finding outliers: statistical-based methods

- Assume a parametric model describing the distribution of the data (e.g., normal distribution)
- Apply a statistical test that depends on:
  - ▶ Data distribution
  - ▶ Parameter of distribution (e.g., mean, variance)
  - ▶ Number of expected outliers (or confidence limit)





# Finding outliers: statistical-based methods

- Example: Univariate Outlier detection

$Age = \{3, 56, 23, 39, 156, 52, 41, 22, 9, 28, 139, 31, 55, 20, -67, 37, 11, 55, 45, 37\}$

- Statistical parameters are:

$$\mu = 39.9$$

$$\sigma = 45.65$$

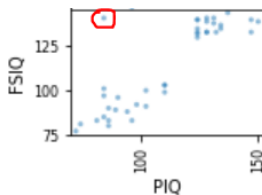
- If we select that the threshold value for normal distribution of data is

$$Threshold = \mu \pm 2\sigma$$

- then all data out of range  $[-54.1, 131.2]$  will be potential outliers:  
 $\{156, 139, -67\}$

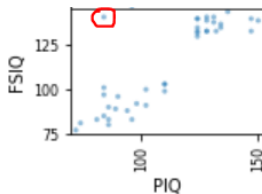
# Finding outliers: statistical-based methods

- Example: Bivariate Outlier detection



# Finding outliers: statistical-based methods

- Example: Bivariate Outlier detection



- When correlation between variables and one point is far from the regression line, then it could be an outlier.
- Procedure:
  - 1 Compute regression line
  - 2 Compute standard deviation of errors (distances of points to regression line)
  - 3 Find points that are further than  $k$  times the standard deviation of the regression line

# Finding outliers: Distance-based and Density-based methods

- Some outliers can be found with uni-variate and bi-variate analysis of columns, but some others escape of this analysis.
- Outliers can also be found analyzing rows (instances) instead of columns.
- Key concept is that one example is one outlier when its *far away of the other examples* or is located in a *zone of low density*.
- Two kind of methods:
  - ▶ **Distance approach.** An instance  $o$  in a data set  $S$  is an outlier if the its  $k$ -th nearest neighbor is at a too high distance when compared with rest of examples of  $S$ .
  - ▶ **Density approach:** An instance  $o$  in a data set  $S$  is an outlier if at least a fraction  $p$  of the samples in  $S$  lies at a distance greater than  $d$